



National Textile University
Department of Computer Science

Subject

Operating System

Submitted to:

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Registration Number

23-NTU-CS-1128

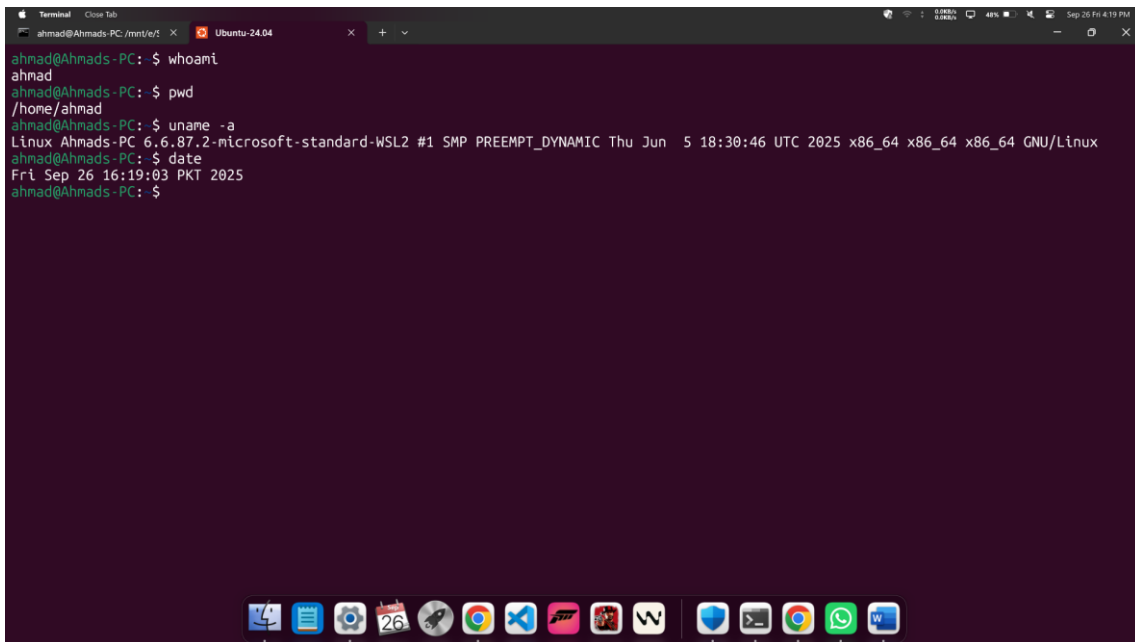
Lab No.

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Semester

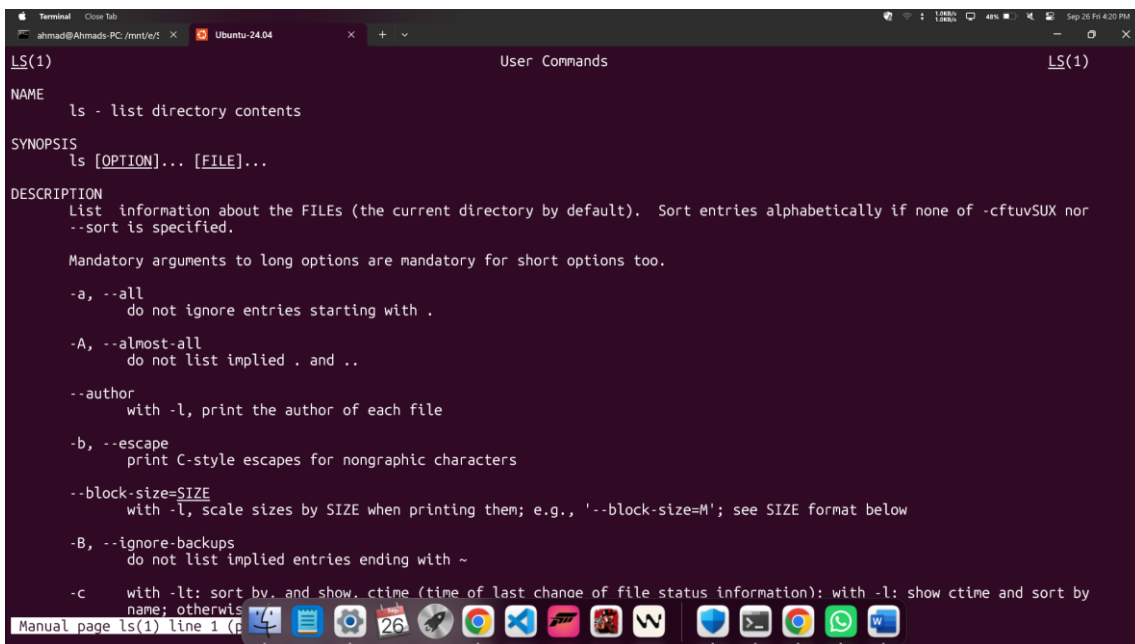
5th

1.1: Basic Commands:



```
ahmad@Ahmads-PC: /mnt/e/ $ whoami
ahmad
ahmad@Ahmads-PC: /mnt/e/ $ pwd
/home/ahmad
ahmad@Ahmads-PC: /mnt/e/ $ uname -a
Linux Ahmads-PC 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 5 18:30:46 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
ahmad@Ahmads-PC: /mnt/e/ $ date
Fri Sep 26 16:19:03 PKT 2025
ahmad@Ahmads-PC: /mnt/e/ $
```

1.2: Getting Help in Linux:



```
LS(1) User Commands LS(1)
NAME
ls - list directory contents
SYNOPSIS
ls [OPTION]... [FILE]...
DESCRIPTION
List information about the FILES (the current directory by default). Sort entries alphabetically if none of -cftuvSUX nor
--sort is specified.

Mandatory arguments to long options are mandatory for short options too.

-a, --all
do not ignore entries starting with .
-A, --almost-all
do not list implied . and ..
--author
with -l, print the author of each file
-b, --escape
print C-style escapes for nongraphic characters
--block-size=SIZE
with -l, scale sizes by SIZE when printing them; e.g., '--block-size=M'; see SIZE format below
-B, --ignore-backups
do not list implied entries ending with ~
-c
with -lt: sort by, and show, ctime (time of last change of file status information); with -l: show ctime and sort by
name; otherwise
Manual page ls(1) line 1 (p)
```

```
Terminal Close Tab
ahmad@Ahmads-PC: /mnt/e/  x Ubuntu-24.04  x + v
ahmad@Ahmads-PC: $ man ls
ahmad@Ahmads-PC: $
ahmad@Ahmads-PC: $ ls --help
Usage: ls [OPTION]... [FILE]...
List information about the FILES (the current directory by default).
Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.

Mandatory arguments to long options are mandatory for short options too.
-a, --all                do not ignore entries starting with .
-A, --almost-all        do not list implied . and ..
--author                 with -l, print the author of each file
-b, --escape             print C-style escapes for nongraphic characters
--block-size=SIZE        with -l, scale sizes by SIZE when printing them;
                        e.g., '--block-size=M'; see SIZE format below

-B, --ignore-backups     do not list implied entries ending with ~
-c                       with -lt: sort by, and show, ctime (time of last
                        change of file status information);
                        with -l: show ctime and sort by name;
                        otherwise: sort by ctime, newest first

-C                       list entries by columns
--color[=WHEN]           color the output WHEN; more info below
-d, --directory          list directories themselves, not their contents
-D, --dired              generate output designed for Emacs' dired mode
-f                       list all entries in directory order
-F, --classify[=WHEN]   append indicator (one of */=>@|) to entries WHEN
                        likewise, except do not append '*'
--file-type              across -x, commas -m, horizontal -x, long -l,
                        single-column -1, verbose -l, vertical -C

--full-time              like -l --time-style=full-iso
-g                       like -l. but do not list owner
--group-directories-first
```

```
Terminal Close Tab
ahmad@Ahmads-PC: /mnt/e/  x Ubuntu-24.04  x + v
ahmad@Ahmads-PC: $ which ls
/usr/bin/ls
ahmad@Ahmads-PC: $ type ls
ls is aliased to `ls --color=auto'
ahmad@Ahmads-PC: $
```

2.1: Linux directory Structure:

```
Terminal Close Tab
ahmad@Ahmads-PC: /mnt/te/5 x Ubuntu-24.04 x + v
ahmad@Ahmads-PC: $ ls /
bin boot etc init lib usr-is-merged lost+found mnt proc run/sbin usr-is-merged srv tmp var
ahmad@Ahmads-PC: $ ls -la
total 48
drwxr-x--- 6 ahmad ahmad 4096 Sep 26 16:20 .
drwxr-xr-x 3 root root 4096 Sep 19 00:09 ..
-rw-r----- 1 ahmad ahmad 1881 Sep 26 15:04 .bash_history
-rw-r--r-- 1 ahmad ahmad 220 Sep 19 00:09 .bash_logout
-rw-r--r-- 1 ahmad ahmad 3771 Sep 19 00:09 .bashrc
drwx----- 3 ahmad ahmad 4096 Sep 19 14:23 .cache
-rw-r--r-- 1 ahmad ahmad 62 Sep 25 23:46 .gitconfig
-rw-r----- 1 ahmad ahmad 20 Sep 26 16:20 .lessht
drwxr-xr-x 3 ahmad ahmad 4096 Sep 26 00:03 .local
-rw-rw-r-- 1 ahmad ahmad 0 Sep 26 14:28 .motd_shown
-rw-r--r-- 1 ahmad ahmad 807 Sep 19 00:09 .profile
drwx----- 2 ahmad ahmad 4096 Sep 25 23:55 .ssh
-rw-r--r-- 1 ahmad ahmad 0 Sep 19 00:10 .sudo_as_admin_successful
drwxr-xr-x 5 ahmad ahmad 4096 Sep 19 14:23 .vscode-server
ahmad@Ahmads-PC: $ cd /hoe
-bash: cd: /hoe: No such file or directory
ahmad@Ahmads-PC: $ cd /home
ahmad@Ahmads-PC: /home$ cd ~
ahmad@Ahmads-PC: $ cd -
/home
ahmad@Ahmads-PC: /home$
```

2.2: Basic Navigation:

```
Terminal Close Tab
ahmad@Ahmads-PC: /mnt/te/5 x Ubuntu-24.04 x + v
ahmad@Ahmads-PC: /home$ pwd
/home
ahmad@Ahmads-PC: /home$ ls
ahmad
ahmad@Ahmads-PC: /home$ ls -l
total 4
drwxr-x--- 6 ahmad ahmad 4096 Sep 26 16:20 ahmad
ahmad@Ahmads-PC: /home$ ls -la
total 12
drwxr-xr-x 3 root root 4096 Sep 19 00:09 .
drwxr-xr-x 22 root root 4096 Sep 26 14:27 ..
drwxr-x--- 6 ahmad ahmad 4096 Sep 26 16:20 ahmad
ahmad@Ahmads-PC: /home$ ls -lh
total 4.0K
drwxr-x--- 6 ahmad ahmad 4.0K Sep 26 16:20 ahmad
ahmad@Ahmads-PC: /home$ cd
ahmad@Ahmads-PC: $ cd ..
ahmad@Ahmads-PC: /home$ cd
ahmad@Ahmads-PC: $ cd ~
ahmad@Ahmads-PC: $ cd /
ahmad@Ahmads-PC: $
```

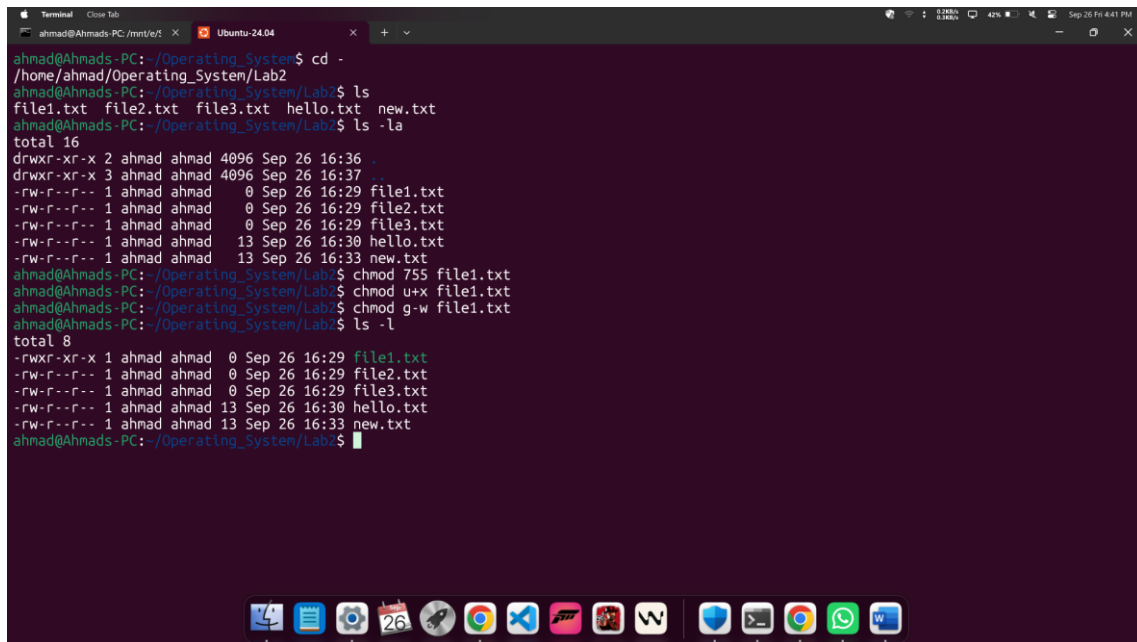
3.1: Linux Directory Operations:

```
Terminal Close Tab
ahmad@Ahmads-PC: /mnt/ro/ x Ubuntu-24.04 x + v
ahmad@Ahmads-PC: $ mkdir Operating_System
ahmad@Ahmads-PC: $ mkdir Operating_System/Lab2
ahmad@Ahmads-PC: $ cd
.cache/ .local/ .ssh/ .vscode-server/ Operating_System/
ahmad@Ahmads-PC: $ cd Operating_System/Lab2/
ahmad@Ahmads-PC: ~/Operating_System/Lab2$ touch file1.txt
ahmad@Ahmads-PC: ~/Operating_System/Lab2$ touch file2.txt file3.txt
ahmad@Ahmads-PC: ~/Operating_System/Lab2$ nano hello.txt
ahmad@Ahmads-PC: ~/Operating_System/Lab2$ echo "hello Linux!" > hello.txt
ahmad@Ahmads-PC: ~/Operating_System/Lab2$ cat hello.txt
hello Linux!
ahmad@Ahmads-PC: ~/Operating_System/Lab2$ less hello.txt
ahmad@Ahmads-PC: ~/Operating_System/Lab2$ head hello.txt
hello Linux!
ahmad@Ahmads-PC: ~/Operating_System/Lab2$ tail hello.txt
hello Linux!
ahmad@Ahmads-PC: ~/Operating_System/Lab2$ wc hello.txt
 1  2 13 hello.txt
ahmad@Ahmads-PC: ~/Operating_System/Lab2$
```

3.2: Copying , Moving and Deleting:

```
Terminal Close Tab
ahmad@Ahmads-PC: /mnt/ro/ x Ubuntu-24.04 x + v
ahmad@Ahmads-PC: ~/Operating_System/Lab2$ cp hello.txt backup.txt
ahmad@Ahmads-PC: ~/Operating_System/Lab2$ cd ..
ahmad@Ahmads-PC: ~/Operating_System$ cp -r Lab2 my_Lab_Backup
ahmad@Ahmads-PC: ~/Operating_System$ ls
Lab2  my_Lab_Backup
ahmad@Ahmads-PC: ~/Operating_System$ cd /
bin/   etc/   lib64/  opt/   sbin/   sys/
bin.usr-is-merged/ home/  lost+found/  proc/   sbin.usr-is-merged/ tmp/
boot/  lib/   media/  root/  snap/   usr/
dev/   lib.usr-is-merged/ mnt/   run/   srv/    var/
ahmad@Ahmads-PC: ~/Operating_System$ cd -
/home/ahmad/Operating_System/Lab2
ahmad@Ahmads-PC: ~/Operating_System/Lab2$ mv backup.txt new.txt
ahmad@Ahmads-PC: ~/Operating_System/Lab2$ cd ..
ahmad@Ahmads-PC: ~/Operating_System$ rm -r my_Lab_Backup
ahmad@Ahmads-PC: ~/Operating_System$ ls
Lab2
ahmad@Ahmads-PC: ~/Operating_System$
```

4.1: Understanding File Permissions:



A terminal window titled 'Terminal' with a close button. The window shows a user named 'ahmad' at a machine named 'Ahnads-PC' in the directory '/mnt/te/5'. The user navigates to '/home/ahmad/Operating_System/Lab2' and lists files: 'file1.txt', 'file2.txt', 'file3.txt', 'hello.txt', and 'new.txt'. Then, the user runs 'ls -la' showing detailed permissions for the directory and files. The permissions for the files are: 'file1.txt' (-rw-r--r--), 'file2.txt' (-rw-r--r--), 'file3.txt' (-rw-r--r--), 'hello.txt' (-rw-r--r--), and 'new.txt' (-rw-r--r--). The user then runs 'chmod 755 file1.txt', 'chmod u+x file1.txt', and 'chmod g-w file1.txt'. Finally, the user runs 'ls -l' showing the updated permissions for 'file1.txt' as '-rwxr-xr-x'.

```
ahmad@Ahnads-PC: /mnt/te/5$ cd -  
/home/ahmad/Operating_System/Lab2  
ahmad@Ahnads-PC: /home/ahmad/Operating_System/Lab2$ ls  
file1.txt file2.txt file3.txt hello.txt new.txt  
ahmad@Ahnads-PC: /home/ahmad/Operating_System/Lab2$ ls -la  
total 16  
drwxr-xr-x 2 ahmad ahmad 4096 Sep 26 16:36 .  
drwxr-xr-x 3 ahmad ahmad 4096 Sep 26 16:37 ..  
-rw-r--r-- 1 ahmad ahmad 0 Sep 26 16:29 file1.txt  
-rw-r--r-- 1 ahmad ahmad 0 Sep 26 16:29 file2.txt  
-rw-r--r-- 1 ahmad ahmad 0 Sep 26 16:29 file3.txt  
-rw-r--r-- 1 ahmad ahmad 13 Sep 26 16:30 hello.txt  
-rw-r--r-- 1 ahmad ahmad 13 Sep 26 16:33 new.txt  
ahmad@Ahnads-PC: /home/ahmad/Operating_System/Lab2$ chmod 755 file1.txt  
ahmad@Ahnads-PC: /home/ahmad/Operating_System/Lab2$ chmod u+x file1.txt  
ahmad@Ahnads-PC: /home/ahmad/Operating_System/Lab2$ chmod g-w file1.txt  
ahmad@Ahnads-PC: /home/ahmad/Operating_System/Lab2$ ls -l  
total 8  
-rwxr-xr-x 1 ahmad ahmad 0 Sep 26 16:29 file1.txt  
-rw-r--r-- 1 ahmad ahmad 0 Sep 26 16:29 file2.txt  
-rw-r--r-- 1 ahmad ahmad 0 Sep 26 16:29 file3.txt  
-rw-r--r-- 1 ahmad ahmad 13 Sep 26 16:30 hello.txt  
-rw-r--r-- 1 ahmad ahmad 13 Sep 26 16:33 new.txt  
ahmad@Ahnads-PC: /home/ahmad/Operating_System/Lab2$
```